

Cyanobacteria as generators of reactive oxygen species – A review of analysis methods and case studies

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ABSTRACT

Reactive oxygen species (ROS) are omnipresent in the environment and serve as important modulators of geochemical cycles. Phytoplankton is the major producer of ROS in water. Cyanobacteria, on one hand, are the most sensitive component of phytoplankton to ROS; on the other, they are producers of ROS with ecological roles that are still not fully understood. The correlation of blooms with increased ROS levels necessitates a deep understanding of the mechanisms responsible for ROS production by cyanobacteria. Reliable analytical methods are thus needed. This review begins with the presentation of ROS as a factor in water biochemistry and the interactions between ROS and cyanobacteria. It then presents reports on ROS analysis following cyanobacteria exposure to physical and chemical stressors using fluorescence probes. The possibilities for selective and qualitative analysis of ROS were critically discussed. In the conclusion section, the main drawbacks of the methods based on fluoresceine-based fluorescence probes, challenges to overcome and future needs are highlighted.

1. Introduction

Reactive oxygen species (ROS) can be defined as short-lived molecules containing oxygen atoms with redox state between O_2 and H_2O , following reduction reactions [1,2].

- (1) $O_2 + e^- \rightarrow O_2^-$ (superoxide – e.g. via photoreduction)
- (2) $O_2^- + e^- + 2H^+ \rightarrow H_2O_2$ (hydrogen peroxide - e.g. via photochemical reaction with dissolved organic matter - DOM)
- (3) $Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH^- + OH^\bullet$ (hydroxyl radical via Fenton or Haber-Weiss Reaction)
- (4) $HO^\bullet + H^+ + e^- \rightarrow H_2O$

ROS have half-lives ranging from nanoseconds to hours in aqueous environments [1–3]. In a normal metabolic process in aerobic cells, ROS arises mainly from oxygen reduction during respiration. In cyanobacteria and other photosynthetic organisms, photosynthetic electron transport serves as an additional ROS source [4]. This process also generates reactive nitrogen species (RNS), and there is significant crosstalk between ROS and RNS [5]. Certain ROS levels play essential roles in metabolic regulation, while elevated ROS levels can damage cellular components. Since most ROS, aside from H_2O_2 , cannot easily diffuse or be actively transported across membranes, cells have evolved

both enzymatic and non-enzymatic defense mechanisms to manage oxidative stress [6]. Enzymatic defenses include catalases, superoxide dismutase, and peroxidases, while non-enzymatic defenses involve glutathione, vitamins A, C, and E, as well as carotenoids. The commonly known state of oxidative stress occurs when the balance between oxidants and antioxidants is disrupted. The sources and regulation of oxidative stress in cyanobacteria are comprehensively reviewed by Lafiti et al. [4]. Environmental stressors, such as high light, increased temperature, salt stress, nutrient limitations, and chemical toxicants, are known inducers of oxidative stress in cyanobacteria [7,8]. The impact of pollution from anthropogenic chemicals on oxidant-antioxidant balance in cyanobacteria remains one of the least explored areas of research.

Oxidative stress is probably a factor in cyanobacteria bloom limitation and cell apoptosis, which is further connected with the presence of cyanotoxins in water [4]. Another issue is ROS released outside the cells, as such molecules can act as antibiotics or signaling agents with a potentially underappreciated role in ecosystems. For example, the production of extracellular ROS was proven for filamentous dinitrogen-fixing cyanobacteria in the genus *Trichodesmium* [9]. The cyanobacteria are found to be especially sensitive toward ROS in comparison to other phytoplanktonic taxa [9,10], which has already been used in natural water reservoirs for controlling blooms [2]. On the other hand, cyanobacteria can produce external ROS to improve iron

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absorption. This suggests that ROS could influence the structure and function of cyanobacterial communities, and the increased number of cyanobacterial blooms can accelerate ROS concentration in water. ROS plays a key role in the aquatic oxygen cycle and actively interacts with other geochemical cycles [2,11]. Therefore, tracking ROS production by cyanobacteria is crucial for understanding environmental ecology and assessing the broader implications of oxidative processes in aquatic ecosystems. Comprehensive studies on oxidative stress in cyanobacteria are needed to understand, among other things, the bloom cycle and the effects of climate change and anthropogenic pollution on primary producers.

The invention of fluorescence methods in 1990's enable tracking nanomoles of H_2O_2 in surface water [2]. Currently, the primary technique for intracellular and extracellular ROS detection is based on fluorescence probes activated by oxidation. However, recent research has shown that many of the methods used so far are not selective enough to distinguish between ROS types [12]. Furthermore, fluorescence probes' performance was never verified for cyanobacterial cells. To date, no comprehensive review of the application of fluorescence probes for cyanobacterial ROS tracking has been conducted. Therefore, the aim of this review is to present a possibility in order to analyze intracellular and extracellular ROS produced by cyanobacteria and critically discuss the qualitative and quantitative powers of obtained methods. We summarize the methods for ROS analysis under laboratory conditions focused on cyanobacteria as emerging microorganisms. The critical discussion aims to highlight the limitations and advantages of fluorescence probes for ROS detection. Furthermore, case studies on the impact of physical and chemical stressors, with special attention to anthropogenic pollutants, are presented.

2. Comparison of selected species of ROS produced by cyanobacteria

Reactive oxygen species (ROS) produced by cyanobacteria differ in both their intracellular functions and environmental roles.

Singlet oxygen ($^1\text{O}_2$) is primarily generated in photosystem II (PSII) due to the excitation of chlorophyll by light. Upon light absorption, chlorophyll reaches a triplet excited state ($^3\text{Chl}^*$), which can transfer energy to molecular oxygen, forming $^1\text{O}_2$ - a highly reactive and damaging form of oxygen. Within cyanobacterial cells, $^1\text{O}_2$ can oxidize chlorophyll molecules, damage PSII proteins, and attack membrane lipids, resulting in the degradation of the photosynthetic machinery.

Despite its toxicity, $^1\text{O}_2$ also plays a role in cellular signaling - a phenomenon referred to as the dual role of ROS. It triggers repair mechanisms in PSII, especially by promoting the replacement of the D1 protein, a key component of the PSII reaction center [13]. It also activates stress-responsive genes such as *psbA* (encoding D1) and *ocp*, which encodes the Orange Carotenoid Protein (OCP). OCP mitigates ROS generation by reducing energy transfer from the light-harvesting phycobilisomes to photosystems, thus limiting the formation of $^3\text{Chl}^*$ and subsequently $^1\text{O}_2$ [14].

Superoxide anion (O_2^-), though less reactive than other ROS, is a major precursor for more harmful species. It is mainly produced in photosystem I under high light stress [4]. Due to its charge, O_2^- cannot diffuse across lipid cell membranes and acts near its production site, where it damages thylakoid membranes and releases Fe^{2+} from iron-sulfur clusters, disrupting key enzymatic processes (Fig. 1). Importantly, O_2^- is also part of a redox-dependent signaling system that responds to light stress. It is sensed by the PrqR-CalA regulatory system, which activates the sigma factors SigB and SigD. These sigma factors promote transcriptional responses that enhance the expression of protective genes and contribute to the oxidative stress defense network [4, 15].

The released Fe^{2+} from O_2^- damaged proteins can participate in the Fenton reaction, leading to the formation of the highly reactive hydroxyl radical ($\text{OH}^\bullet$). This reaction is preceded by the conversion of O_2^- to

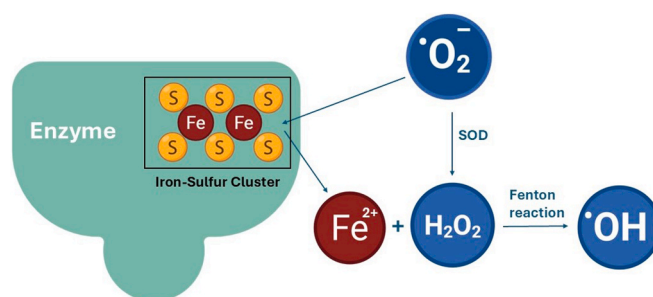


Fig. 1. Formation of the hydroxyl radical from superoxide anion in cyanobacterial cells.

hydrogen peroxide (H_2O_2) via superoxide dismutase (SOD). H_2O_2 , though less reactive than $\text{OH}^\bullet$, is membrane-permeable and can exert oxidative effects in both the thylakoid and cytoplasmic compartments.

Cyanobacteria counteract H_2O_2 through specialized defense systems, including PerR and the histidine kinase Hik33, which differ from regulatory mechanisms in heterotrophic bacteria like *E. coli* [4,16]. However, the complete network of ROS defense in cyanobacteria remains incompletely understood, and additional regulatory components are likely involved.

ROS, especially $\text{OH}^\bullet$, pose significant internal threats to nucleic acids, proteins, and lipids. Radical $\text{OH}^\bullet$ is extremely short-lived and reacts immediately at its site of formation. In DNA, it causes strand breaks and base modifications, leading to mutations [17]. Oxidative damage to mRNA can impair translation and protein synthesis [18]. Lipid peroxidation, particularly targeting unsaturated fatty acids, compromises membrane integrity and can generate further ROS, including $^1\text{O}_2$. Proteins are similarly vulnerable: $\text{OH}^\bullet$ indiscriminately modifies side chains and backbones, while O_2^- and H_2O_2 selectively damage iron-containing proteins like ferritin [17]. Even essential enzymes, such as the β -subunit of ATP synthase, can suffer oxidative damage, impairing core cellular functions.

In addition to their intracellular effects, ROS produced by cyanobacteria also play roles in ecological interactions, including allelopathy, ichthyotoxicity, and iron mobilization. In some cases, they may even promote growth in microbial communities [19].

3. The interdependence between ROS and phytoplankton dynamics

ROS are the components of the oxygen cycle in the environment. Nevertheless, full knowledge of spatial and temporal occurrence of ROS in water is lacking. The primary sources of ROS in water are phytoplankton and chromophoric dissolved organic matter (CDOM) [2]. Only a few studies have been conducted to determine the natural temporal and seasonal levels of ROS in water. Morris et al. [2] provide a comprehensive review of H_2O_2 concentrations across diverse aquatic environments globally, reporting values ranging from 2 nM to 4.5 μM . Notably, H_2O_2 concentrations in rainwater are substantially higher than those in surface waters, ranging from 1 to 85 μM , primarily due to photochemical reactions occurring in the atmosphere. Seasonal variations significantly influence H_2O_2 dynamics. Yoon et al. [20] demonstrated that summer H_2O_2 concentrations can be up to three times higher than winter levels, highlighting the role of temperature and irradiance in oxidative processes. Also, Cory et al. [21] reported that temporal fluctuations in H_2O_2 concentrations were strongly associated with seasonal bloom dynamics in lacustrine ecosystems. In natural aquatic systems, ROS are often undetectable due to their rapid reactions with dissolved organic matter and metals. Consequently, most reported ROS concentrations are derived from controlled laboratory experiments, which may not fully capture their dynamics in complex environmental matrices.

Phytoplankton presence is intricately linked to ROS dynamics in

natural waters, as these organisms both generate and respond to ROS. Review of reactive species production by photosystem II (PSII) as a response to high light and high temperature can be found in literature [7]. The primary phytoplankton-derived ROS are H_2O_2 and O_2^- . H_2O_2 is generated both intracellularly and through cell surface reactions, while O_2^- is produced during intracellular metabolic processes, as mentioned in previous section. However, O_2^- has an extremely short half-life (seconds to minutes) and does not diffuse outside the cell, as it is rapidly converted into H_2O_2 by superoxide dismutase (SOD). The extracellular O_2^- is a process of surface activity of enzymes. The concentrations ($\text{fmol cell}^{-1} \text{ hr}^{-1}$) of ROS produced by selected species across phytoplankton can be found in review [19].

Despite its ecological significance, the role of extracellular ROS in phytoplankton physiology remains incompletely understood. Hypothesized ecotoxicological functions include ichthyotoxicity, allelopathy, growth regulation, and iron speciation, but these effects have only been documented in selected species and cannot be generalized across all phytoplankton. It's important to note that ROS can have beneficial effects on the environment [22]. They serve as important cell signaling molecules [23] that participate in organismal protective responses and may regulate resistance to pathogens. Additionally, ROS participate in redox reactions with trace metals (Fe, Cu, Mn), enhancing nutrient bioavailability, which supports phytoplankton growth and influences carbon and oxygen cycles [2,22]. Extracellular ROS production has been implicated as a toxic mechanism in harmful algal species, such as *Chattonella marina* [24], and various others (review in [25]). However, the rate of extracellular ROS production varies among species, as highlighted in a review by Diaz and Plummer [19]. In addition to species-specific differences, factors such as growth stage, cell density, cell lysis, light intensity, temperature, salinity, and iron availability further regulate extracellular ROS production.

Exposure to ROS in the environment can induce oxidative stress in phytoplankton, influencing their physiological responses [2]. Cyanobacteria, which are particularly sensitive to ROS, can experience oxidative damage from light-irradiated CDOM [10,20]. Additionally, environmental stressors such as temperature fluctuations, acidification, and contaminants (e.g., pesticides, pharmaceuticals, and other toxic substances) can further exacerbate oxidative stress in phytoplankton [2, 26]. ROS can interfere with redox reactions involving trace metals, thereby altering the bioavailability of essential nutrients crucial for phytoplankton growth [2]. Studies have shown that H_2O_2 can affect the speciation of iron, rendering it unavailable to phytoplankton and consequently inhibiting their growth [27]. The extracellular reduction of Fe^{3+} has been observed in *Microcystis aeruginosa*, enhancing iron uptake. This process is mediated by organic exudates, which, upon irradiation, induce the generation of O_2^- , facilitating Fe^{3+} reduction [28]. One of the first cyanobacteria to be subjected to experiments testing growth inhibition under the influence of H_2O_2 was *Prochlorococcus*, a cyanobacterium that inhabits temperate waters and tropical oligotrophic oceans. It has been found that *Prochlorococcus* relies on heterotrophic bacteria for increased tolerance to H_2O_2 [29,30]. In the absence of these bacteria, oceanic ROS levels can rise, rapidly inhibiting cyanobacterial growth. *Prochlorococcus* is a particularly sensitive cyanobacterial species to ROS, possibly due to the lack of enzymes responsible for defense against H_2O_2 in its genome. While other cyanobacterial species are generally less sensitive than *Prochlorococcus*, they are still more susceptible to ROS compared to many other aquatic organisms, as reviewed by Morris et al. [2]. Unicellular cyanobacteria of the genus *Synechocystis* [31], *M. aeruginosa* [32] and the filamentous cyanobacteria *Trichormus variabilis* [32] showed higher sensitivity (mortality already at exposure to $10 \mu\text{M}$ H_2O_2 [2]) compared to green algae such as *Pseudokirchneriella subcapitata* and diatoms like *Navicula semimulum* [32], which survived exposure up to 1 mM H_2O_2 [2].

Researchers suggest that environmental ROS concentrations are generally lower than the toxicity thresholds observed in laboratory studies of aquatic organisms [33]. However, this assumption requires

careful verification, as culture media used in laboratory experiments may contain unrecognized ROS sources, potentially leading to artificially elevated resistance levels in cultivated microorganisms [1,34]. This discrepancy complicates assessments of wild aquatic organisms' true sensitivity to ROS in their natural habitats. Furthermore, accurately quantifying ROS levels in environmental samples remains challenging due to technical limitations in current analytical methods, further hindering direct comparisons between laboratory and field conditions.

4. Methods of extracellular and intercellular ROS analysis

There are direct and indirect methods for quantification of ROS in natural waters [35], and in water in general [3,36]. Various techniques can be applied for ROS and RNS analysis, including fluorescence, luminescence, spectrophotometry, high-performance liquid chromatography (HPLC) with UV-Vis detection, gas chromatography, electron spin resonance, nuclear magnetic resonance, iodine titration, and capillary electrophoresis [5]. According to the latest literature, the fluorescent methods has been used for cyanobacteria (Tables 1 and 2). In general, probes react with various forms of ROS to produce a product that exhibits fluorescence and can provide spatial and temporal information about cellular systems [36] (Fig. 2).

Some probes penetrate directly into cells, while others detect ROS in the environment surrounding the cells. Like other ROS detection methods, fluorescent probes vary in selectivity. However, due to the short-lived and highly reactive nature of ROS, most fluorescent probes are considered non-selective, meaning they primarily indicate overall ROS levels rather than distinguishing between specific ROS types.

To improve accuracy, combining multiple detection methods can provide a more reliable assessment of both intercellular and intracellular ROS distribution. In addition, the use of appropriate, controlled, and reproducible procedures, such as conducting positive controls (with ROS donors), negative controls (with ROS scavengers), proper sample washing, maintaining constant staining/scanning times within a set of experiments, and using optimal pH and turgor pressure, can contribute to accurate results. It is also critical to find the optimal experimental conditions: incubation time, temperature, or probe concentration, as these values vary significantly between different model phototrophic organisms (higher plants, algae, and cyanobacteria), fungi, or "fungus-like" organisms (oomycetes) [57].

In general, unicellular structures, like pollen, green algae, and thin-walled spores can be stained easily [57]. However, in the case of cyanobacteria, effective staining requires a higher probe concentration and an extended incubation time to achieve optimal results. For example, in the case of the unicellular cyanobacterium *Synechocystis*, a widely used photosynthetic model, the Singlet Oxygen Sensor Green (SOSG) concentration is usually increased from the commonly used $50 \mu\text{M}$ – $250 \mu\text{M}$, along with an increase in incubation temperature from room temperature to 37°C [58]. This adjustment is necessary due to the cyanobacterial cell wall's higher cross-linking of polysaccharides and the production of an external mucoid sheath. The average cell wall thickness is 10 nm in unicellular species and 15 – 35 nm in filamentous species. An extreme case is *Oscillatoria princeps*, where the cell wall is 700 nm wide [57]. Cyanobacteria are, therefore, challenging species for intracellular ROS analysis. In the paper of Cho et al. [25] there is review of methods for analysis of the ROS in harmful algal bloom (HAB)-forming phytoplankton and it seems that the topic of ROS analysis for algae is more known compared to cyanobacteria.

The *in-situ* detection of extracellular ROS in water environment can be performed by fluorescence-based assays and electrochemical methods [59]. For example, H_2O_2 can be performed by flow injection analysis acridinium ester chemiluminescence. The base is reaction of H_2O_2 with an acridinium ester 10-methyl-9-(*p*-formylphenyl)-acridinium carboxylate trifluoromethanesulfonate. The system was used to determine H_2O_2 concentrations in natural waters, including rain, freshwater, and seawater with detection limit 352

Table 1

The overview of reports on DCFH₂-DA application for reactive oxygen species (ROS) detection in cyanobacteria after exposure to physical and chemical stressors (PAR- photosynthetically active radiation, FCM – flow cytometry, FM - fluorescence microscopy, FS - fluorescence spectrophotometry, F – green fluorescence).

Cyanobacteria Species	Stressor Exposure Time	Stressing Conditions	Cell Density	Probe (final conc.)	Incubation with Stain	Detection Technique	ROS Quantification	Excitation/ Emission [nm]	Ref.
Physical stressors									
<i>Anabaena variabilis</i>	UV-A, UV-B, PAR 12 h	23 °C, shaken regularly	–	DCFH ₂ - DA (5 μM)	1 h (after exposure, shaken)	FM, FS	No	Green F 485/500-600	[37]
<i>Anabaena variabilis</i>	UV-A, UV-B, PAR 24 h	23 °C, shaken regularly	–	DCFH ₂ - DA (5 μM)	1 h (after exposure, shaken)	FM	No	Green F and red F	[38]
<i>Rivularia</i> sp.	UV-A, UV-B, PAR 24 h	23 °C, shaken regularly	–	DCFH ₂ - DA (5 μM)	1 h (after exposure, shaken)	FS	No	Green F 485/520	[39]
<i>Synechococcus elongatus</i> PCC <i>Fremyella diplosiphon</i>	–	–	1 × 10 ⁵ cells mL ⁻¹	DCFH ₂ - DA	1 h (after exposure, shaken, in dark)	FMC	Yes	Green F (emission 515–545 nm) and red F (emission range >670 nm)	[40]
<i>Nostoc</i> sp. strain <i>HKAR-2</i>	UV-A, UV-B, PAR 24 h	25 ± 2 °C, stirred 15–20 rpm	–	DCFH ₂ - DA (5 μM)	1 h (after exposure, shaken, in dark)	FM, FS	No	Green F 488/500-600	[41]
<i>Nostoc</i> sp. strain <i>HKAR-2</i> <i>Scytonema</i> sp.	UV-A, UV-B, PAR 72 h	25 ± 2 °C, stirred 15–20 rpm	optical density of 0.5 at 750 nm	DCFH ₂ - DA (5 μM)	1 h (after exposure, shaken, in dark)	FS	No	Green F 485/500-600	[42]
<i>Fischerella</i> sp. strain <i>HKAR-14</i>	UV-A, UV-B, PAR 72 h	20 ± 2 °C, photoperiod of 14/ 10 h light/dark cycle	–	DCFH ₂ - DA (5 μM)	1 h (after exposure, shaken, in dark)	FM, FS	No	Green F 485/525	[43]
<i>Synechococcus</i> sp.,	Temperature (20, 23 and 26 °C) 96 h	Two-factor design: 20, 23 and 26 °C and salinity 33, 36 and 39 PSU in different configurations	10 ⁴ cells mL ⁻¹	DCFH ₂ - DA	30 min (after exposure, in dark) 10 ⁴ cells·mL ⁻¹	FMC	No	Green F 488/530	[44]
<i>Synechocystis</i> sp.	UV-A, UV-B, PAR (rhythmic light variations) 24 h	photoperiod of 12/ 12 h light/dark cycle	optical density of 0.7 at 750 nm	DCFH ₂ - DA	1 h (after exposure, shaken, in dark)	FS	No	Green F 485/500-600	[45]
<i>Phormidium tenue</i>	UV-B	UV-B and PAR photoperiods	2–5 × 10 ⁴ cells/ mL	DCFH ₂ - DA (5 μM)	1 h (after exposure, shaken, in dark)	FS	No	Green F 485/500-600	[46]
Chemical stressors									
<i>Microcystis aeruginosa</i> <i>Aphanizomenon flos- aquae</i> <i>Anabaena flos-aquae</i> <i>Microcystis aeruginosa</i>	Natural water	–	–	DCFH ₂ - DA	6/12 h	FS	No	Green F 480/530	
<i>Microcystis aeruginosa</i>	Polyphenolic allelochemical pyrogallallic acid (2, 7 and 14 mg L ⁻¹) 72 h	25 ± 2 °C light/dark cycle of 12 h/12 h	–	DCFH ₂ - DA (5 μM)	1 h (after exposure, in dark)	FCM	No	Green F 485/535	[47]
<i>Fischerella</i> sp.	Methyl parathion 96 h	–	–	DCFH ₂ - DA (5 μM)	1 h (after exposure, shaken, in dark)	FM	No	Not specified	[48]
<i>Anabaena sphaerica</i>	Malathion (pesticide) 5, 10, 20 and 30 μg mL ⁻¹ 8 days	28 ± 2 °C, photoperiod of 14/ 10 h light/dark cycle	–	DCFH ₂ - DA (5 μM)	1 h (after exposure, in dark)	FM	No	Not specified	[49]
<i>Synechocystis</i> sp	CNT, CAP, CT 96 h	–	1 × 10 ⁶ cells mL ⁻¹	DCFH ₂ - DA	in the dark at 25 °C for 30 min	FS	No	Green F 485/530	[46]
<i>Microcystis aeruginosa</i> <i>Pseudokirchneriella subcapitata</i>	Allelochemicals 4 h	Room temperature, light/dark conditions, static	5 × 10 ⁶ cells mL ⁻¹ 2 × 10 ⁶ cells mL ⁻¹	DCFH ₂ - DA (5 μM)	1 h (before exposure)	FS	No	Green F 485/530	[50]
<i>Anabaena fertilissima</i>	250 mM NaCl, NaCl + Na ₂ SO ₄ (125 mM) 12 h	28 °C	–	DCFH ₂ - DA (5 μM)	45 min (after exposure)	FM	No	Red F and green F	[51]

(continued on next page)

Table 1 (continued)

Cyanobacteria Species	Stressor Exposure Time	Stressing Conditions	Cell Density	Probe (final conc.)	Incubation with Stain	Detection Technique	ROS Quantification	Excitation/Emission [nm]	Ref.
<i>Rivularia halophila</i>	NaAsO ₂ Na ₃ AsO ₄ ·12H ₂ O 15 days	Stirred	–	DCFH ₂ -DA (5 µM)	1 h (after exposure, shaken, in dark)	FM	No	Green F 492/527	[52]
<i>Synechococcus</i> sp.,	salinity (33, 36 and 39 PSU) 96 h	Two-factor design: 20, 23 and 26 °C and salinity 33, 36 and 39 PSU in different configurations	10 ⁴ cells mL ⁻¹	DCFH ₂ -DA	30 min (after exposure, in dark)	FCM	No	Green F 488/530	[44]

Table 2

The methods based on other than DCFH₂-DCA probe for ROS analysis in cyanobacteria.

Probe	ROS type	Cyanobacteria Stressor	Observed Effect	Ref.
Intracellular Singlet Oxygen Sensor Green (SOSG)	Singlet oxygen	<i>Synechocystis</i> sp. No stressor	Prove that small CAB-like proteins prevented formation of singlet oxygen in the damaged photosystem II complex of the cyanobacterium.	[53]
Extracellular AmplexTM Red	Total ROS	<i>Microcystis aeruginosa</i> , <i>Aphanizomenon flos-aquae</i> , <i>Anabaena flos-aquae</i> and green algae <i>Chlorella</i> sp. heavy metals, antibiotics tetracycline and sulphamethoxazole (1 µg/L)	The higher H ₂ O ₂ level was measured in cyanobacteria culture after antibiotics exposure compared to heavy metals.	[20, 54]
AmplexTM Red	Total ROS	<i>Microcystis aeruginosa</i> <i>Aphanizomenon flos-aqua</i> UV-irradiated natural organic matter	Cyanobacteria are 2 order of magnitude more sensitive to H ₂ O ₂ compared to green algae.	[55]
6-(4-methoxyphenyl)-2-methyl-imidazo [1,2-a] pyrazine-3(7H)-one (MCLA)	Superoxide	<i>Trichodesmium</i> sp. No stressor, natural level measured	Superoxide production rate per cell determined.	[43]
6-(4-methoxyphenyl)-2-methyl-imidazo [1,2-a] pyrazine-3(7H)-one (MCLA) [2-[4-[3,7-dihydro-2-methyl-3-oxoimi-dazo [1,2-a]pyrazin-6-yl] phenoxy] butyramido] ethylamino] sulforhodamine10 (red-CLA)	Superoxide	<i>Trichodesmium erythraeum</i>	Superoxide production rate per cell determined.	[9]
Titanium chloride method	H ₂ O ₂	<i>Phormidium ambiguum</i> combined PAR and nutrient depletion (phosphorus concentrations)	Both PAR regimes and P concentration shortages are shown to enhance H ₂ O ₂ concentrations in cyanobacterial cultures. Protein-related quantification was performed.	[56]

pM in ocean seawater to 58.1 nM in rain [60,61]. The advantages of this method is negligible background fluorescence matrix effects in organic-rich environment. The emitted luminescence at 470 nm can be detected by flow injection analysis using a FeLume system (Waterville Analytical, Waterville, ME), as performer for groundwater [62]. This system was also applied for laboratory analysis of extracellular superoxide of several species of ocean microbes with application of chemiluminescent probe MCLA [63]. MCLA was also used with success for tracking the superoxide in the water column in the eastern equatorial Pacific Ocean [64]. What interesting, the same flow system, but with both probes (acridinium ester, and MCLA) was applied for detection of the extracellular H₂O₂ and O₂⁻ produced by diatoms [65]. The ROS in-situ can be also analyzed by electrochemical methods, for example, scanning electrochemical microscopy (SECM) [66]. For more information we advise to check review of methods for ROS detection in aqueous environments [19,67]. Despite quite well established methods for extracellular ROS analysis, we were unable to found the *in-situ* analysis of ROS produced by cyanobacteria blooms. Such data are obtained for laboratory cultivated phytoplanktonic species [19], with predominance of harmful green algae species.

4.1. Non-selective fluorescence probes for intracellular ROS detection

There are many dihydrocompounds, such as 2',7'-dichlorodihydrofluorescein (DCHF), dihydrorhodamine 123, dihydrorhodamine 6G,

dihydrocalcein, dihydroethidium, 10-acetyl-3,7-dihydroxyphenoxazine (AmplexTM Red) and 2,3,4,5,6-pentafluorodihydrodihydro-4-methylamine, which can be used as fluorescent probes for ROS detection [36]. However, all dihydrocompounds have two major drawbacks. First, it is known that they are originally highly photosensitive compounds, to a greater or lesser degree, and as a result, tend to auto-oxidize, producing high background fluorescence in the absence of ROS during fluorescence measurements that involve excitation light irradiation by laser. Therefore, experimental results obtained with these compounds should be interpreted with caution since the increase in fluorescence intensity is derived from both the presence of ROS and autooxidation by excitation light. Another issue is that these dihydrocompounds have long been widely used as fluorescent probes for the selective detection of H₂O₂. However, as is now known, these probes can react with different types of ROS. Therefore, dihydrocompounds are not suitable for the selective detection of individual ROS; however, they are effective for measuring overall oxidative activity in living cells or tissues.

The derivatives of fluorescein are routinely applied for ROS detection. Among them, 2',7'-dichlorodihydrofluorescein diacetate (DCFH₂-DA) is the most widely used, and due to its high permeability across membranes and cell walls, is used to detect intracellular ROS. Dihydrofluorescein diacetate (DHF-DA) was found to be superior for detecting intracellular oxidants compared with DCFH₂-DA among other fluorescence probes [68]. At the same time, all such probes show no selectivity, and provide an overall response toward oxidative reaction

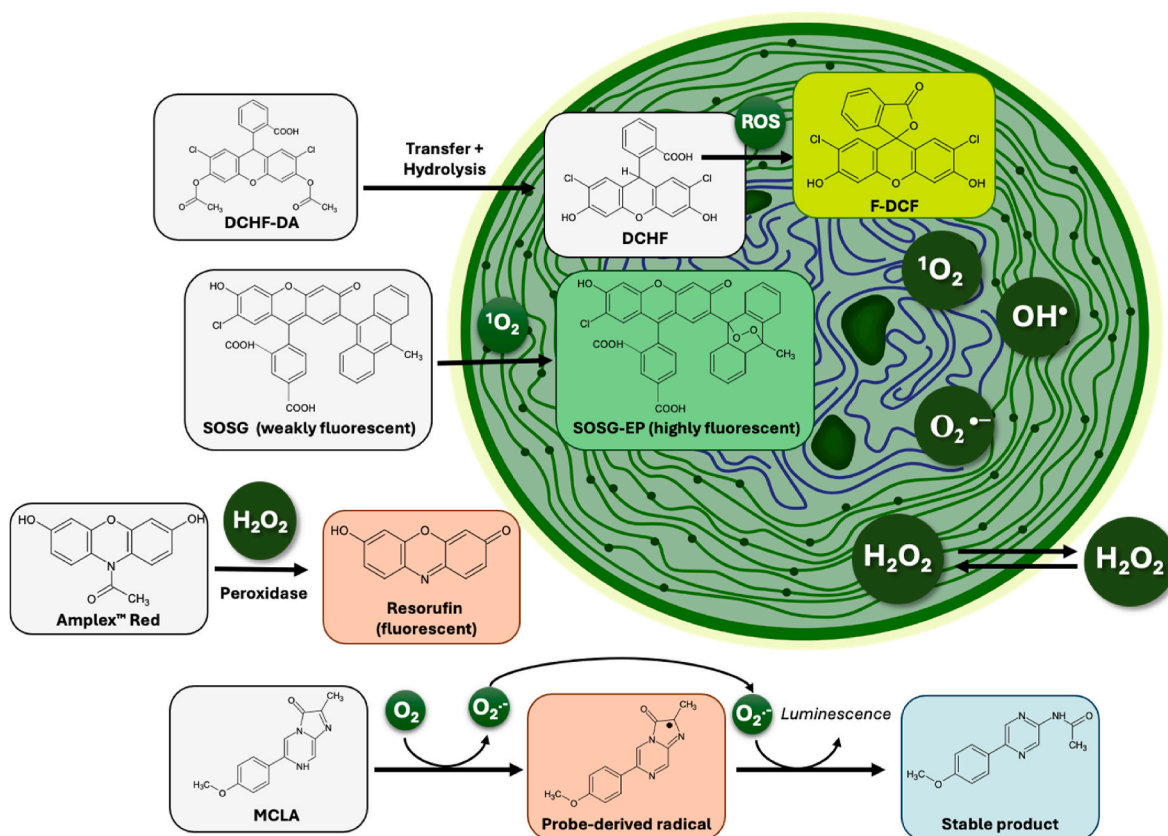


Fig. 2. The fluorescence probes applies for tracking of intracellular and extracellular ROS produced by cyanobacteria.

rather than selective response to single factor. In laboratory tests, the fluorescence was found to constantly grow with incubation time [68, 69], which is also our observation (data not published). DCFH₂-DA assay for analysis the intracellular ROS based on two reaction: first, uptake by cell and deacetylation by esterase to oxidant sensitive species (conversion of DCFH₂-DA to DCHF), and second, oxidation by ROS (DCHF to fluorescent DCF (dichlorofluorescein), named as f-DCF in further text) (Fig. 2) [69]. Besides ROS, reactive nitrogen species (RNS) may also directly or indirectly cause f-DCF formation [37].

Table 1 presents reported studies of cyanobacteria derived intracellular ROS analysis using DCFH₂-DA. Fluorescence spectrophotometry (FS) and fluorescence microscopy (FM) are primarily used as detection techniques, often in combination. In three reports, flow cytometry (FCM) was applied [44,70,71]. Each of these techniques relies on fluorescence detection, and the choice of method in the studies presented in Tables 1 and 2 appears to depend primarily on instrument availability and the specific research objectives. In general, fluorescence-based methods are approximately 1000 times more sensitive than spectrophotometric approaches [72]. FCM, in addition to fluorescence detection, also quantifies cell numbers, while FM provides spatial information about probe distribution on or within cells. Notably, only FS allows for the detection of probe fluorescence in solution, enabling the monitoring of extracellular ROS. To date, no study has directly compared FS, FM, and FCM in the context of ROS analysis. A common feature shared by these techniques is the excitation and emission behavior of the fluorescent probes, as DCF-based probes operate via the same physicochemical mechanisms. Excitation of f-DCF is carried out at wavelengths from 480 nm to 488 nm, while emission is measured at values from 500 nm to 600 nm. Most of the work did not attempt to validate or calibrate the methods used and only determined the overall level of ROS produced by various species of cyanobacteria (and not only) under natural conditions and because of various stressors in controlled laboratory studies.

DCFH₂-DA is particularly applicable to cyanobacteria, as the

structure of their cell wall prevents most probes from penetrating the cell interior. In real-case studies, DCFH₂-DA has been used most frequently (Table 1). Typically, incubation of the test solution with the probe is performed after exposing the cells to a factor that induces oxidative stress, such as UV-A, UV-B, or high photosynthetically active radiation (PAR). The effects of salinity, allelochemicals, pesticides, or carbon nanotubes on oxidative stress in cyanobacterial cells have also been evaluated. DCFH₂-DA is usually used at a final concentration of 5 μM, with a common incubation time of 30–60 min. Since DCFH₂-DA is a light-sensitive fluorescence probe, experiments should be conducted in minimal light. Therefore, incubation is carried out in the dark while shaking the cyanobacteria solution mixed with the probe. The following paragraphs present case study data on the use of DCFH₂-DA in detecting ROS produced by cyanobacterial species, sorted by the factor causing oxidative stress.

A model study type involves determining ROS induced by radiation mimicking solar UV ranges (UV-B, UV-A). For example, in the widely cited work of Rastogi et al. [37], studies were conducted on the cyanobacterial species *Anabaena variabilis*. Both FM and FS were used, obtaining compatible results. The excitation/emission values used were 485/500–600 nm, which are common for the DCFH₂-DA probe. In testing UV-R radiation effects on biochemical changes in cyanobacterial cells (and their surroundings), cut-off filters as monochromators are necessary to determine the accuracy of the wavelength affecting the test organisms. During exposure, the temperature was 23 °C, and the solution was regularly shaken. These conditions are like those used in most studies on this subject. Temperatures typically range from 20 °C [43] to 25 °C [71,41,42], with the solution being stirred or shaken. According to Table 1, UV light exposure time vary from 12 to 72 h [43]. It was observed that DCFH₂-DA was characterized by extensive green background fluorescence due to autooxidation and noticeably increased fluorescence in ROS-generating cells. Post-exposure results revealed that UV-B radiation induced a significant increase in ROS levels, most visible

when a 295 nm cut-off was applied. Additionally, Singh et al. [73] published results on the correlation of rhythmic changes in time and radiation components on ROS production by *Synechocystis* species. The highest ROS concentrations occurred with continuous radiation across the full range of natural light, including UV-B. In 2014, the same team conducted similar research on temporal dynamics of ROS biogenesis using *Anabaena* species and FM [74]. This study used seven different cut-off filters ranging from 280 nm to 400 nm. Images showed no ROS induction during the first 8 h of exposure to simulated solar radiation with different cut-off filters. After 24 h, ROS generation signals were observed in all samples except those kept in the dark. The signals were more pronounced in samples with 320, 305, 295, or 280 nm cut-off filters, indicating the highest ROS generation.

In the same year, Rastogi et al. [37] published a paper that included information on the effects of UV radiation on oxidative stress in *Rivularia* sp. The procedure is similar to the one presented in papers [37], fluorescence spectrophotometry with 485/520 nm emission and excitation, respectively, was used and the cut-off filters were 295 nm, 320 nm and 395 nm. Also in 2017, a full protocol was published for a method of determining ROS in cyanobacteria *Nostoc* sp. strain HKAR-2, where UV-R was the stressor [41]. They showed that solar UV radiation harms cyanobacteria directly by acting on DNA/proteins or indirectly through ROS-induced oxidative damage. As in most tests of this type, samples were taken after 12 h and after 24 h of exposure of cyanobacteria to radiation. A marked increase was observed after 12 h, doubling after 24 h. Similar conclusions were presented for cyanobacteria isolated from the environment *Scytonema* sp. and lab-cultured species *Nostoc* sp. strain HKAR-2 [42], hot spring cyanobacterium *Fischerella* sp. strain HKAR-14 [43].

In 2023, Kholssi et al. [44], investigated temperature and salinity effects on ROS production by *Synechococcus* sp. A temperature range of 20, 23, and 26 °C was investigated. Elevated concentrations of ROS were consistently observed across all tested temperatures, with a slight augmentation noted at the highest temperature, likely due to cell mortality correlating with increased ROS production.

Several papers have used DCFH₂-DA to assess the effects of structurally diverse chemicals on oxidative stress in cyanobacteria, such as allelochemicals [70,50] and pesticides like malathion [49], methyl parathion and arsenic salts [52].

Wang et al. [50] applied DCFH₂-DA to measure the oxidative stress caused by allelochemicals using FS (excitation/emission 485/530 nm). Other analytical parameters also do not differ from those standardly used for DCFH₂-DA (Table 1). However, this work deserves special attention for two reasons. First, it compares the concentration of ROS produced by two different species: cyanobacteria and green algae. Second, an attempt was made to validate the method, which is not usually done, due to the low selectivity of the DCFH₂-DA probe. Validation of the method was performed as a positive control using H₂O₂ to test the reliability of DCFH₂-DA as a probe for detecting intracellular ROS in *M. aeruginosa* and *P. subcapitata*. H₂O₂ significantly increased f-DCF fluorescence generation compared to the respective controls in both cultures. However, the generation of f-DCF fluorescence in *P. subcapitata* cells induced by H₂O₂ was about twice as high as in *M. aeruginosa* at all exposure concentrations over 4 h. The results showed that of all the stressors tested, only (+)-catechin and pyruvic acid could induce ROS formation both in cyanobacteria *M. aeruginosa* and green algae *P. subcapitata* cells. In all of the tested experimental conditions, ROS levels were higher when green algae were used as the test organism. In another work on the influence of allelochemicals on cyanobacteria *M. aeruginosa* [47], FCM technique was used for the intracellular ROS levels measurement with excitation/emission wavelengths of 485/535 nm. Also in this work, a positive correlation was proven between allelochemical pyrogallol acid concentration and intracellular ROS levels, suggesting the autooxidation of pyrogallol acid.

Two studies investigated insecticides' impact on ROS levels in cyanobacteria: Chakraborty et al. [49] tested stressor malathion (192 h of

exposure), and Tiwari et al. [48] tested methyl parathion (MP, 96 h of exposure). In both cases the detection technique applied was FM (Table 1). The cyanobacterial species used were *Anabaena sphearica* [49] and *Fischerella* sp. [48]. It was stated that high dose of malathion (30 mg L⁻¹) induced the generation of ROS. What interesting, the goal of the work of Tiwari et al. [48] was not only to correlate ROS concentration with insecticide concentration, but also to study the MP removal and degradation capacity of this cyanobacterium along with the mechanism it adopted to combat pesticide toxicity at different doses. It was found that at 20 mg L⁻¹ MP, the cyanobacterium effectively modulated its antioxidant defense system, fatty acid and hydrocarbon profiles to promote growth. However, the results obtained after exposing cyanobacterial cells to MP found that the pesticide caused a significant increase in ROS levels in a dose-dependent manner.

Rueda et al. [52] investigated the impact of the presence of NaAsO₂ and Na₃AsO₄·12H₂O across concentrations ranging from 0 to 15,000 mg L⁻¹ on ROS production by *Rivularia halophila* cyanobacterial cells. The measurement of ROS was conducted under conditions typical for DCFH₂-DA (Table 1), apart from the cyanobacteria exposure duration to the stressor, which extended up to 15 days. Initially, an escalation in arsenic salt concentration was accompanied by a gradual rise in ROS concentration, as anticipated. Furthermore, their findings elucidated that the bioreduction of As(V) to As(III) emerged as the principal transformation process, simultaneously diminishing levels of antioxidants and protective mechanisms aimed at ROS mitigation.

In the paper by Swapnil et al. [51], the influence of salinity (250 mM) on ROS production by *Anabaena fertilissima* was determined using DCFH₂-DA. Salt-treated cells showed a distinct biphasic accumulation of ROS. The first phase in ROS accumulation lasted 2 h, the second generation, less intense, was observed for from the 4th hour of salt stress to 12th hour of exposure. The biphasic timing of ROS accumulation corresponded to the response of cyanobacterial cells to salt stress and may serve as a potential trigger for an early defense response. Also, Pathak et al. [42] presented results on the effect of salinity (up to 250 mM) on oxidative stress in cyanobacteria from the species *Scytonema* sp. and *Nostoc* sp. strain HKAR-2. ROS level peaked after 60 h of exposure to salinity and then began to decrease. Possible causes were cell death caused by the extremely high salinity or early defense response. The mentioned works differ in the detection technique used: in the case of Swapnil et al. [51] is FM, while Pathak et al. [42] used FS. The study conducted by Kholssi et al. [44] also aimed to elucidate the impact of salinity on ROS production by cyanobacteria, employing FCM for analysis. However, no definitive correlations were discerned across the spectrum of salinity investigated (33–39 PSU), except for the consistent increase in ROS generation in the presence of salt. Various exposure durations were also employed, spanning 12, 72, and 96 h, alongside a range of temperatures from 20 to 28 °C.

There is also an interesting paper published in 2021 on measuring oxidative stress in different species: cyanobacteria *M. aeruginosa*, *Aphanizomenon flos-aquae*, *Anabaena flos-aquae* and green algae *Chlorella* sp. in water coming directly from the natural environment [20]. The detection was performed using FS under standard conditions (excitation/emission: 480/530 nm). Controls were performed for all selected species under commonly used conditions: under fluorescent light (4000 lx) with a cycle consisting of 14 h of light and 10 h of darkness at 20 ± 1 °C, stirred every 8 h. The composition of the medium was adjusted to reflect the composition of the environmental water under study. In addition to DCFH₂-DA, the AmplexTM Red was applied for extracellular ROS detection. Unfortunately, it is not clear if both reagents were added to same samples or measurement was performed in separate samples and how the samples were treated.

There is also data in the literature on the effect of carbon nanotubes (CNTs) on the toxic effects of chloramphenicol and tetracycline on the cyanobacterium *Synechocystis* sp. [75]. Both ROS quenching experiments and proteomic analysis were performed. FS was employed to detect ROS after staining by DCFH₂-DA. The results showed that

ROS-induced oxidative stress contributed to the toxicity of CNTs.

As was presented, the staining with DCFH₂-DA can be detected by FM, FS of FCM. The FCM is an alternative tool for ROS analysis in cyanobacteria analysis and possess several advantages. In a study of Mondal et al. [71] the FCM protocol utilized single-celled cyanobacteria, *Synechococcus elongatus*, and the filamentous *Fremyella diplosiphon*. They employed an FCM-based method to measure ROS in a large population of cyanobacterial cells efficiently. The authors rightly highlight the limitations of FM, which is restricted to small sample sizes due to time-consuming processes and background noise from dye photooxidation. Additionally, FM struggles with visualizing ROS in small, spherical cells. The study proposes FCM-based ROS monitoring as a solution, offering high-throughput capabilities and the simultaneous recording of various cellular parameters. However, unlike other detection methods, the FCM-based method lacks complete validation. The protocol includes optimization of the DCFH₂-DA dye concentration to ensure maximal ROS detection while minimizing cell death. Practical advice for using FCM with cyanobacterial cells is provided, emphasizing the need for careful sonification to disperse filamentous cyanobacteria or cell aggregates. Nonetheless, not all species may be suitable for this treatment.

Technically, the staining with the DCFH₂-DA is easy to perform and the price of reagent relatively low. The DCFH₂-DA can be prepared in DMSO [50] or ethanol [37,43,42,74,51]. In most cases, ROS are measured by DCFH₂-DA by *in vivo* techniques, what mean that appropriate mass of staining reagent is added directly to treated cultures (sometimes to washed pellet). The experiment can also be performed in reverse way, by firstly loading cells with stain, then performing treatment. As example, Wang et al. [50] performed test by pre-incubation of cells with DCFH₂-DA in PBS buffer, then cells were washed and resuspended in respective culture medium, and after adjustment of cell density (2-5 × 10⁶ cells mL⁻¹) the allelochemical was added. The control was the cells with only DCFH₂-DA; solvent control (DMSO) was also performed. The background fluorescence formation was determined with the mixture of BG-11 + growth medium, cells without loading the DCFH₂-DA and different concentrations of the allelochemicals. Obviously, in a case of PAR or UV as stressor only *in vivo* studies can be performed, because the DCFH₂-DA is light-unstable.

What was noticed after reports analysis, that the majority of works did not present a way of negative or positive controls performance. The presence of stress was measured by comparison of response of treated and not-treated cultures. DCFH₂-DA is a probe widely used for ROS detection in various cell types over several decades. However, its use comes with certain limitations: A. the oxidized fluorescent product of DCHF is membrane-permeable and can leak out of cells over time; B. incomplete DCF trapping may complicate data interpretation and hinder precise evaluation of intracellular oxidation; C. some cells have low esterase activity; D. the exact species responsible for oxidation is unknown [69].

4.2. Non-selective AmplexTM red for extracellular ROS determination

AmplexTM Red (10-acetyl-3,7-dihydroxyphenoxazine) reagent is used for non-selective determination of ROS. However, due to its inability to penetrate cell membranes/walls, it only allows the determination of extracellular ROS. The advantages of its use are simple procedures, fast reaction, and commercial availability of full experimental setups. Additionally, this reagent offers stability, generates less background noise, and produces a red-fluorescent product that is more easily detected than similar reduced methylene blue derivatives. In the presence of horseradish peroxidase (HRP), AmplexTM Red reacts with H₂O₂ in a 1:1 stoichiometry to form highly fluorescent resorufin (Fig. 2). However, it should be remembered that this represents a simplification, and AmplexTM Red is also capable of reacting with other ROS. The reference to DCFH₂-DA or AmplexTM Red for the determination of H₂O₂ content is since H₂O₂ is the most stable form of ROS. However, the ability of these probes to interact not only with various forms of ROS, but

also with RNS, cannot be neglected.

This simplification was still widely used until 2015. A paper from 2014 by Leunert et al. (2014) indicated the possibility of AmplexTM Red reagent application determining H₂O₂ produced by cyanobacteria, but the H₂O₂ concentrations were very low compared to those obtained using green algae. In fact, AmplexTM Red is more commonly used for green microalgae [76,77]. FS as detection technique for AmplexTM Red is used with the following excitation and emission values used: 530/590 nm [20], and 572/585 nm [10] (Table 2). In the work of Leunert et al. [10], the Amplex Red was used to quantify the ROS produced by UV-irradiated natural organic matter (NOM), and calibration was performed using six H₂O₂ levels ranging between 10⁻⁹ and 10⁻⁶ M. Unfortunately, in the work [20] no specific parameters of AmplexTM method were given. The authors stated that the analysis of ROS was performed both by AmplexTM and DCFH₂-DA, and test heavy metals, antibiotics tetracycline and sulphamethoxazole (1 g L⁻¹) as stressor. The higher H₂O₂ level was measured in cyanobacteria culture after antibiotics exposure compared to heavy metals.

Despite its advantages, AmplexTM Red has not been widely used for the determination of extracellular ROS, and the reasons for this remain unclear. One possible explanation is that its use has been largely overshadowed by other probes, such as DCFH₂-DA. In addition, while Amplex offers excellent sensitivity and stability, its reliance on horseradish peroxidase (HRP) to react with H₂O₂ can introduce variability under different experimental conditions. This may limit its wider application.

4.3. Selective Singlet Oxygen Sensor Green for intracellular singlet oxygen determination

Singlet Oxygen Sensor Green (SOSG) reagent was used to visualize singlet oxygen (¹O₂) production inside cells [78]. This reagent is also based on fluorescence, like DCFH₂-DA or AmplexTM Red. However, unlike other available fluorescent and chemiluminescent singlet oxygen detection reagents, SOSG does not show any appreciable response to other ROS, including hydroxyl radical, superoxide and nitric oxide [79]. Its limitation is unequal penetration between cell types (especially limited transport through mucoid sheath of cyanobacteria) and photo-sensitization [78]. This probe shows weak blue fluorescence with excitation peaks at 372 and 393 nm and emission peaks at 395 and 416 nm. In contrast, in the presence of ¹O₂ it produces SOSG endoperoxide (SOSG-EP), which emits green fluorescence like fluorescein (excitation/emission maxima ~504/525 nm) (Fig. 2). The SOSG reagent can degrade over time in some solutions and can become fluorescent at alkaline pH in the absence of singlet oxygen. Nevertheless, with proper control, the intensity of the green-fluorescent signal can be correlated with the concentration of singlet oxygen, without significant interference from other ROS. However, quantitative analysis of ¹O₂ formation performed exclusively based on SOSG endoperoxide fluorescence images is misleading [78].

Sinha et al. [58] used SOSG to determine ¹O₂ in cells of cyanobacteria of the *Synechocystis* species (Table 2). However, in this case, the stressor was not used to induce oxidative stress in cyanobacteria, and the goal was to see if small CAB-like proteins prevented formation of ¹O₂ in the damaged PSII complex. Cells were stained with SOSG and kept at 37 °C for 15 min to increase the efficiency of fluorochrome uptake by the mucous envelope surrounding the cyanobacterial cell wall. The samples were then incubated for 3 h at room temperature in the dark or with photosensitizing exposure. Suspensions of cells in deuterium oxide and histidine served as positive and negative controls, respectively. SOSG may be easily photosensitized during incubation and microscopy [57]. Therefore, to minimize artifacts, it is recommended that the sample be stained in the dark and visualized by a confocal laser scanning microscope. Based on the results obtained, the presence of ¹O₂ was indicated, and the role of small CAB-like proteins in stress reduction proven (most probably by the binding of free chlorophyll released from the damaged

PSII complexes) [58]. No other current publications on the use of SOSG against cyanobacteria were found, but it was used for diatoms [79].

4.4. Chemiluminescence probe for extracellular superoxide determination

Hansel et al. proposed a method based on the absorption of radiation produced by chemiluminescence [40] by adapting the method of [9]. This method was used to measure extracellular O_2^- production by the cyanobacterium *Trichodesmium*. The chemiluminescent probe used was 6-(4-methoxyphenyl)-2-methyl-imidazo [1,2-a]pyrazine-3(7H)-one (MCLA), which is a Cypridina luciferin analog [80] (Fig. 2). In the absence of a natural bioluminescent protein, MCLA generates chemiluminescence in response to superoxide production by the cells [40]. A possible mechanism for the chemiluminescence of MCLA was described in work by Zielonka et al. [81].

The advantage of this probe is that it does not sensitize to O_2^- production and therefore does not lead to false-positive results. Its pH optimum for luminescence generation is closer to the physiological near-neutral range [82]. However, MCLA-dependent chemiluminescence is enhanced by naturally occurring NADH in the cell, and since this electron donor is present in most cells, the true release of O_2^- can be masked [83]. In the research by Hansel et al. [40] (Table 2) extracellular O_2^- was measured by passing filtered seawater (AFSW) through filter-supported *Trichodesmium* colonies directly into the FeLume Mini system (two separate fluid lines, one for the analyte solution and the other for the MCLA reagent), where it was mixed with MCLA. This method, unlike most ROS detection methods, was fully calibrated, allowing the quantification of O_2^- in the test solutions. For calibration, standard solutions of potassium dioxide (KO_2) in NaOH (pH 12.5) were prepared with the addition of 90 μ M diethylene triamine pentaacetic acid (DTPA) to capture trace impurities that could otherwise significantly shorten the peroxide lifetime. Superoxide concentrations in the primary standards were quantified by measuring the difference in absorbance at 240 nm before and after the addition of superoxide dismutase. The secondary standard was the aged, filtered solutions amended with DTPA. The limit of detection for these measurements was 100 ± 7 pM of O_2^- . The rate of O_2^- production by populations sampled in the Sargasso Sea was normalized to the total number of *Trichodesmium* colonies or *Coscinodiscus* cells added to provide normalized rates (final pmol units of colonies per hour). As the conclusion, was stated that *Trichodesmium* colonies produce O_2^- independent to light presence and are a key ROS source in the surface of oligotrophic ocean. MCLA assay was applied to track O_2^- in various species of harmful algal blooms [25]. Considering the lack of calibration or validation of the methods used in most already presented work, the validation performed in the described work is impressive.

5. Findings

This review highlights several key points associated with current detection methods for extracellular and intracellular ROS in cyanobacteria, summarized in Fig. 4, but they can be extended to other cell types.

5.1. The non-selectivity of ROS detection methods

At the outset, it is necessary to emphasize the already known facts of using DCFH₂-DA as a fluorescence probe for ROS. Unfortunately, in contradiction with the generally repeated opinion, DCFH₂-DA does not react directly with H_2O_2 but requires a peroxidase or transition metal catalyst [84,85] and can produce H_2O_2 itself in intermediate reactions ('self-fulfilling prophesy') [86]. Furthermore, the relationship between the intensity of f-DCF fluorescence and the concentration of the ROS is often nonlinear [87]. It can be an effect of artifactual amplification of the fluorescence intensity via a redox-cycling mechanism involving an intermediate radical, DCF[•], because of several one-electron-oxidizing species oxidize DCHF to f-DCF, as presented in work by Bonini et al.

[88] and by Wardman et al. [85]. In addition, cytochrome c, a heme protein that is released from mitochondria to the cytosol during apoptosis, is capable of oxidizing DCHF directly or indirectly via peroxidase-type mechanism, forming f-DCF [89]. In the work of Chen et al. [90] more than 20 more interferences for f-DCF formation are reviewed. Therefore, even if differences in DCF fluorescence between control and exposed/stressed cells are observed, one must consider the various strictures listed above for proper observation interpretation. As a conclusion, DCFH₂-DA probe cannot be reliably used to measure intracellular H_2O_2 and other ROS [91]. The only useful information from the DCHF-DA probe is the exploration of redox signaling mechanisms in cells. This probe, despite its widespread use, primarily provides qualitative data (relative differences) rather than precise quantitative measurements. This fact is often overlooked in microalgae studies. The other weaknesses of the fluorescent probes, mainly DCFH₂-DA, were the topic of several previous reviews [90,91,91–93].

5.2. False-positive and false-negative results

There are more reactive species in cells than ROS and RNS [92,93] for which we researchers do not have tools for measuring in cyanobacteria. What is essential is that most available methods for ROS analysis in aqueous solutions lack specificity. They do not find specific ROS but rather show a general response to oxidative stressors. Many fluorescence probes work based on non-specific oxidation and therefore do not distinguish between O_2^- , H_2O_2 and OH^- , even though these species have different biological effects. For example, presumably DCFH₂-DA is used to detect H_2O_2 , but it also reacts with peroxynitrite ($ONOO^-$) [91], and other cellular ROS [94], leading to false-positives results. The non-selectivity of ROS detection methods threatens understanding of ROS dynamics in natural environments, potentially leading to misinterpretation of their ecological roles. Non-selectivity also affects experimental design, data interpretation, and reproducibility. One key issue may be overestimating or underestimating the impact of ROS on phytoplankton communities. For example, if DCFH₂-DA is used without controlling selectivity, leading researchers to falsely conclude that cyanobacteria are under oxidative stress, when in fact other factors (e.g., nutrient deficiency) may be at work. AmplexTM Red is used to quantitatively measure H_2O_2 , but its signal is affected by metal ions or organic matter, which can lead to an underestimation of extracellular ROS.

Interpretation of ROS measurements is often inadequate, especially with chemical stressors that can interfere with the detection process. Chemical stressors can interfere with ROS detection assays, leading to false-negative results. For example, some chemicals can inhibit deacetylation of the DCFH₂-DA probe, leading to underestimation of oxidative stress [95]. Cellular antioxidants present in biological samples can scavenge ROS, reducing their availability to react with detection probes and leading to underestimation of ROS levels, as highlighted by review [96].

Furthermore, natural components such as DOM, salinity and pH fluctuations also affect the performance of ROS probes, leading to false-negative or false-positive results. For example, SOSG is widely used to detect singlet oxygen, but high DOM concentrations in freshwater ecosystems quench fluorescence, leading to underestimation of singlet oxygen levels. Therefore, a conclusion could be that photochemical ROS production is negligible in some lakes, when in fact singlet oxygen is present but undetectable due to DOM interference [97]. Many ROS-sensitive probes show pH-dependent fluorescence [98]. Deviations from the optimal pH can alter the behavior of the probe, leading to inaccurate detection of ROS [55]. discussed that changes in environmental pH can interfere with ROS detection, causing misleading results. What can be done is to use proper buffering agents to support a constant pH level during experiments, ensuring stable probe performance.

Two other factors that can lead to false-negative results are the chemical reactivity of the chemical components present in the test solution and nanoparticle interference. Both factors can cause quenching

or enhancement of fluorescence signals (so both can also cause false positives) [56,99]. In addition, some compounds can generate ROS when exposed to light, which can lead to artifactual results in ROS assays, as indicated by ref. [100] in the case of pharmaceuticals, e.g. quinine, tamoxifen, ketoprofen, diclofenac and many other. In addition, in buffered phytoplankton culture media there is continuous H_2O_2 production, causing sublethal chronic exposure to cyanobacteria [1]. Similarly - redox-active metals, e.g. Fe^{2+} can induce oxidation of the probe. False-positive results can be also obtained by artifact generation. For example, in the case of DCFH₂-DA the intermediate radical form during oxidation can lead to superoxide production and finally more H_2O_2 [95]. In the case of AmplexTM Red enzymatic interference can be a big obstacle regarding getting correct results [101,102]. Because the probe functioning depends on horseradish peroxidase (HRP) presence, endogenous reducing agents like NADH and glutathione (GSH) can interfere. Therefore, they can go auto-oxidation or HRP-mediated oxidation and generate H_2O_2 .

5.3. Lacks in ROS detection studies

Current studies (Tables 1 and 2) predominantly focus on unicellular cyanobacteria, while filamentous types remain underrepresented. This gap is significant, as filamentous cyanobacteria contribute to ROS production differently than their unicellular counterparts. Additionally, a cell type in general plays a crucial role in the consistency and reliability of ROS measurements. Research has shown that ROS probe behavior can vary between different cell types, and in some cases, extracellular probes may be taken into the cell interior, potentially altering the accuracy of ROS detection (e.g. in case of SOSG) [103].

There is a notable absence of simultaneous quantification of extracellular and intracellular ROS in the studies reviewed. Such studies can be performed, for example, for mouse cells [104]. This dual quantification is essential for a comprehensive understanding of the role of cyanobacteria in the total ROS in the environment. Fig. 3 summarizes the fluorescence channels used to detect intracellular and extracellular ROS in cyanobacterial cultures. The most common approach involves using stains such as DCFH₂-DA and AmplexTM Red, whose fluorescence

differs from the natural autofluorescence of chlorophylls and phycobilins found in cyanobacteria. Only a few studies utilizing fluorescence microscopy have specifically analyzed the natural red fluorescence of cyanobacteria [105]. None of the work track autofluorescence of phycobilins. Tracking the natural autofluorescence of cyanobacterial cells (Fig. 3) together with synthetic probes can provide deeper insights into the stress responses of cyanobacteria, and ROS evolving. The development of methods for the simultaneous quantification of extracellular and intracellular ROS is essential to obtain a complete assessment of the oxidative environment. This is particularly important for cyanobacteria, whose mechanisms of ROS production remain insufficiently understood.

5.4. Validation improvements and limits of detections

There are technical needs for ROS detection performance. Rigorous validation of ROS detection methods is essential to ensure reliable and reproducible results under different experimental conditions. The use of positive and negative controls, such as H_2O_2 or other ROS donors and scavengers, is crucial for the standardization of these methods. A reference protocol using DCFH₂-DA could facilitate the comparison of stressors, strains, and experimental factors. This protocol should specify reference cell concentrations for unicellular cyanobacteria and biomass measurements for filamentous forms. In addition, quantification and normalization of ROS should be performed in a more standardized way, such as reporting median fluorescence per cell or per biomass unit. The establishment of universal standardized measurement protocols adapted to aquatic environments would enhance the comparability, reproducibility, and transparency of data.

Some of the publications presented in Tables 1 and 2 attempted to quantify the ROS content of water produced by cyanobacteria. In principle, DCFH, AmplexTM Red and other ROS probes (e.g. titanium chloride – Table 2) allow some degree of quantification of ROS in surface water samples. However, there are important limitations to be aware of. Firstly, DCFH₂-DA and AmplexTM Red react not only with H_2O_2 , but also with $\cdot\text{OH}$, $\text{ONOO}\cdot$, $\text{NO}\cdot$ and other ROS. None of the samplers presented in this paper provide information about which specific ROS were involved in the reaction; instead, they only indicate the total amount of

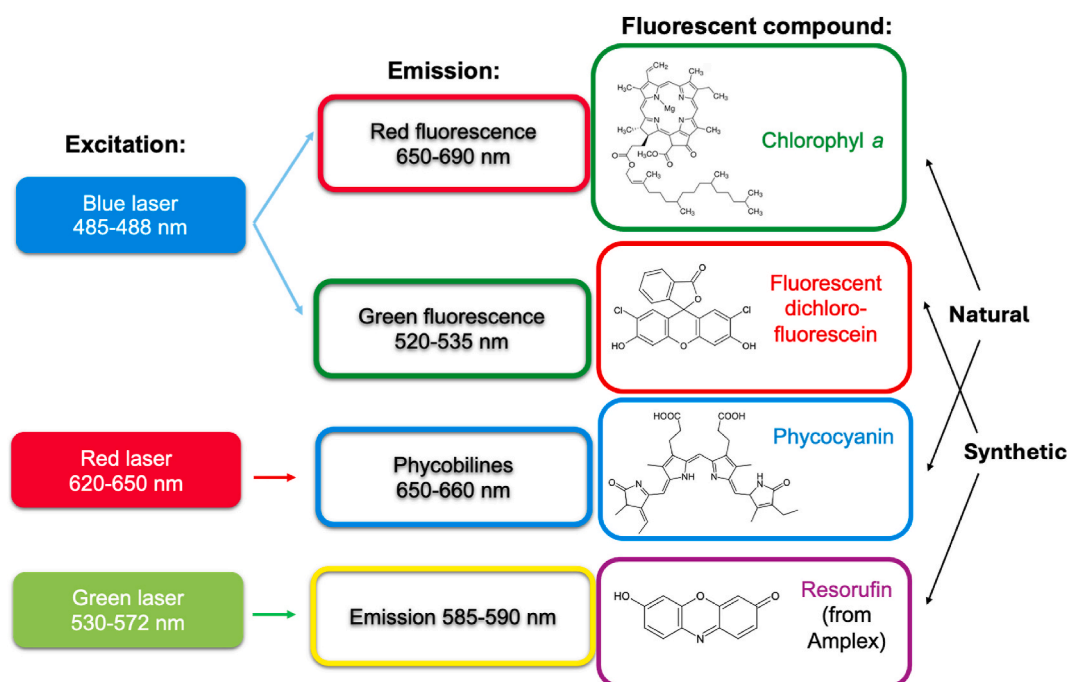


Fig. 3. Canals of fluorescence of naturally occurred substances in cyanobacteria (chlorophyll a, phycobilins) and stains - dichlorofluorescein for intracellular ROS detection and resorufin (product of AmplexTM Red) for extracellular ROS detection.

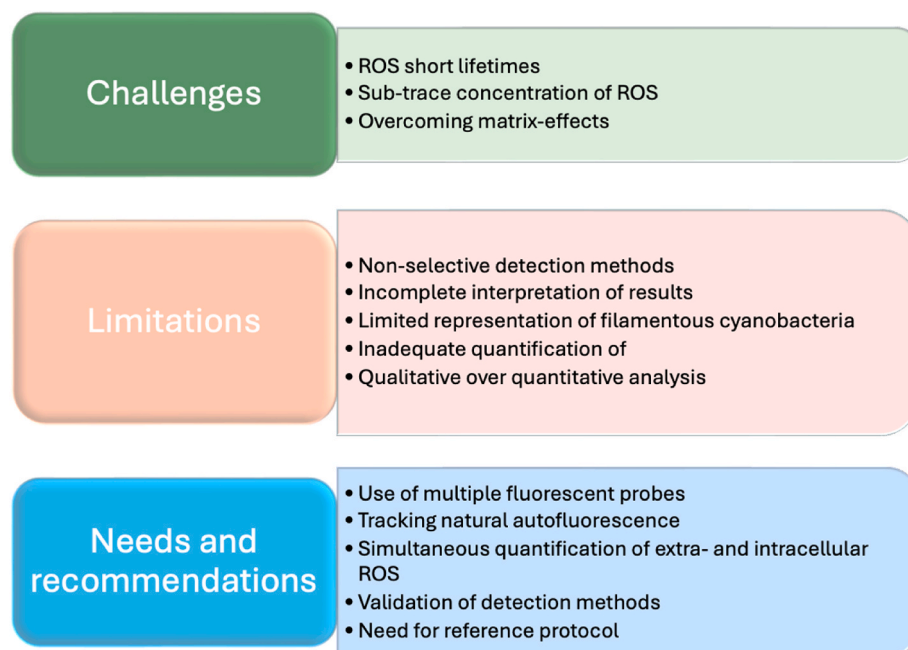


Fig. 4. The summary of challenges of ROS detection, the limitations of available methods and the recommendation for future studies for tracking of ROS production by cyanobacteria.

reactive species that oxidized the probe. In other words, the measurement is quantitative but non-selective, representing the aggregate reactivity of all ROS (i.e., total ROS) (Table 3).

AmplexTM Red also requires catalysts like peroxidase (HRP), which complicates conversions. Most samplers also auto-oxidize easily (especially with light/heat), which increases the fluorescence background, and the effect of pH, oxygen levels and metal ions on the efficiency of the probes - can interfere with the linearity and repeatability of measurements. Thus, the matrix of natural waters (e.g. DOM, metals, organic pollutants) can mask, generate or absorb ROS, requiring standardization of methods and daily calibration (e.g. H₂O₂), background control and avoidance of light exposure. Several research papers have proposed possible solutions to this situation (Table 2), such as: H₂O₂ per unit of protein (μmol/g) [106], fluorescent RFU/MFI, converted by calibration to ROS [71] (Table 1), comparing fluorescence intensity between treatments [46]. These methods use titanium chloride [106], DCFH₂-DA + cytometry, with calibration against H₂O₂ [71,46] and DCFH₂-DA + spectrofluorometry, with calibration by relative quantification [46] allowing results to be presented in quantitative units Table 1.

Studies of reactive oxygen species (ROS) produced by cyanobacteria - whether qualitative assessments, observations of concentration

changes, or the limited number of quantitative analyses - have all been conducted under laboratory conditions, including those using natural water samples (Table 1). This is due to the extreme instability of ROS, which typically persist for only seconds to minutes in aqueous environments. Accurate quantification therefore requires that samples be processed immediately after collection. The available literature reports only laboratory-based measurements of ROS in cyanobacteria, most often employing fluorescent probes such as DCFH-DA or AmplexTM Red [71,41]. To date, no validated methodologies exist for the direct, quantitative measurement of ROS released by cyanobacteria in field conditions. As a result, rapid sample collection and transport under controlled conditions - low temperature and protection from light - are essential to minimize ROS degradation and ensure reliable analytical results.

The graphical relative comparison of probes in order of sensitivity, selectivity, quantification potential and applicability for intra- and extracellular ROS detection is presented on Fig. 5.

6. Conclusions and perspectives

The ROS plays a critical role in both biogeochemical cycles and the

Table 3
Comparison of ROS samplers used in measuring oxidative stress of cyanobacteria.

Probe	ROS Detected	Selectivity	Sensitivity	Detection Method	Quantification	Intracellular Use	Extracellular Use	Used In
DCFH ₂ -DA (2',7'-Dichlorodihydrofluorescein diacetate)	General ROS (e.g., H ₂ O ₂ , •OH, ROO•, ONOO ⁻)	Non-selective	Moderate (~μM range)	Fluorescence (~485/530 nm)	Semi-quantitative	Yes (esterase-dependent)	Possible fluorescent leakage	[37, 41]
Amplex Red + HRP	General ROS (dedicated to H ₂ O ₂)	Low-moderate (HRP-aided, not specific)	High (1–10 nM)	Fluorescence (~560/590 nm)	Semi-quantitative	No	Yes	[40]
SOSG (Singlet Oxygen Sensor Green)	Singlet oxygen (¹ O ₂)	Moderate-high (selective in vitro)	Moderate (~nM, but qualitative)	Fluorescence (~504/525 nm)	Qualitative	Weak penetration in cyanobacteria	No	[37, 53]
MCLA (Methoxy-luminol analog)	Superoxide (O ₂ ⁻ , extracellular only)	Good (with SOD control)	High (100–500 pM)	Chemiluminescence	Quantitative (with KO ₂ calibration)	No	Yes	[40]

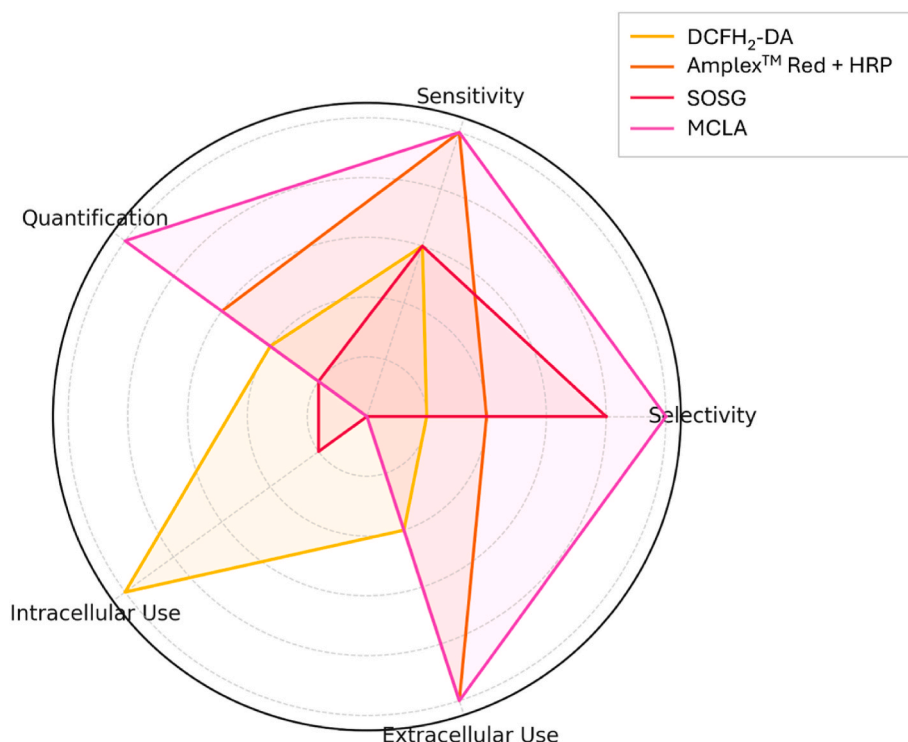


Fig. 5. Relative comparison of probes in order of sensitivity, selectivity, quantification potential and applicability for intra- and extracellular ROS detection for cyanobacterial cells.

modulation of phytoplankton communities within aquatic environments. However, the comprehensive understanding of factors influencing ROS concentrations in natural water bodies remains incomplete. Monitoring of cyanobacterial-derived ROS in natural waters is infrequent, with most data derived from controlled laboratory studies. One of the limitations is the lack of verified methods for tracking intracellular and extracellular ROS, as presented in this review. In practice, the methods track changes in ROS level after stress with comparison to control what is not quantitative information.

As the conclusion of this review, we present the following advice, and perspectives. Improving methodologies for ROS detection is crucial due to the key role of ROS in geochemical cycles. Current methods, while valuable, have significant limitations that need to be addressed to improve the accuracy and reliability of ROS measurements. The limits of detection of ROS by fluorescence probes need to be tracked. A change of perspective on the DCFH₂-DA test result is necessary because this probe does not react with H₂O₂, or even ROS directly as presented in Section 4. By adopting more comprehensive approaches scientists can gain deeper insights into the dynamics of cyanobacterial ROS and their impact on aquatic ecosystems. Of various selective methods to tract ROS (presented in reviews [3,94,96]) only a few were tested for cyanobacteria. Summary of recent advances of detection methods and probes for tracking ROS in biological systems can be found in refs. [72,107,108]. Numerous new selective probes and detection methods are available, which can be tested for ROS detection in phytoplankton. Therefore, studies on the application of new probes are urgent. In addition, studies need to be extended to filamentous cyanobacteria. The simultaneous measurement of natural and synthetic fluorescence of cells could be beneficial for strengthened statements. The next step in this research should be to investigate the impact of anthropogenic factors and climate change on ROS production by cyanobacteria.

CRediT authorship contribution statement

Hanna Lis: Writing – review & editing, Writing – original draft,

Investigation. **Klaudia Kwidzińska:** Writing – review & editing, Writing – original draft, Visualization, Investigation. **Katarzyna Bethke:** Writing – review & editing, Writing – original draft, Investigation. **Magda Caban:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve grammar and logical sequence in the article. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Magda Caban reports financial support was provided by National Science Centre Poland. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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